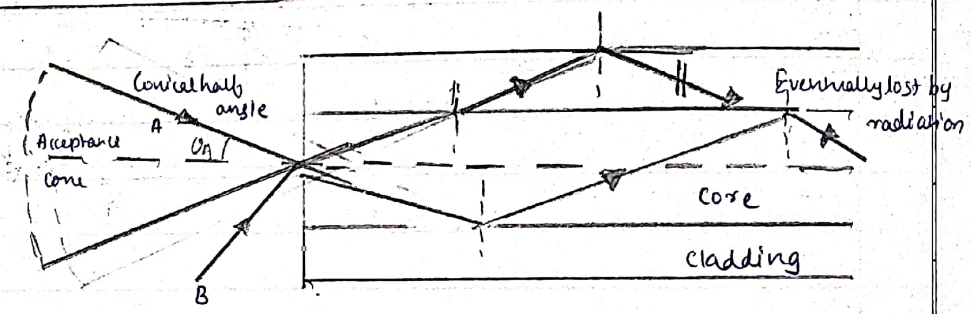


Fig 1: Schematic of optical fibre both in cross section and lateral views.

Fig 2: The geometry of launching a light ray is shown ~~in Fig 2.~~



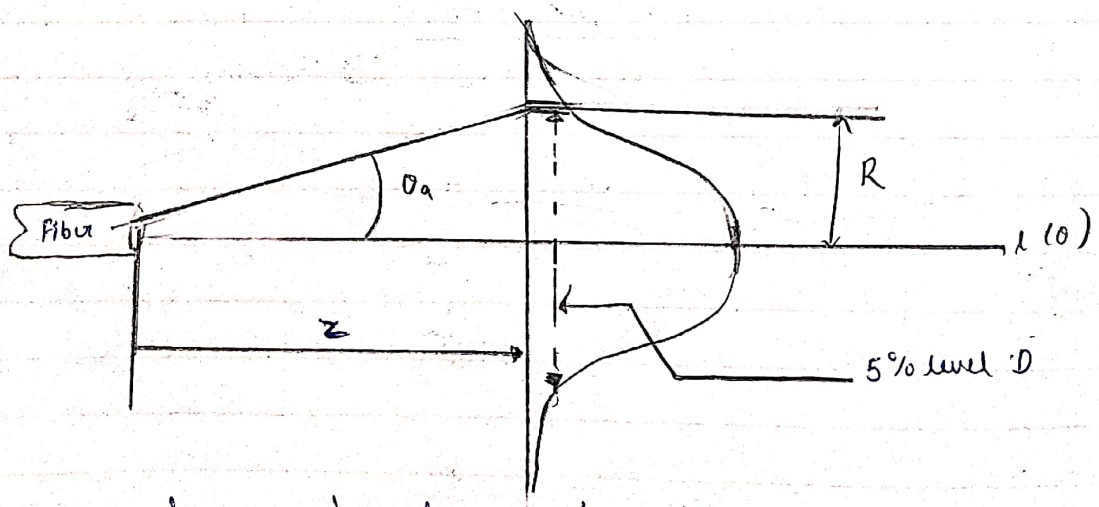


Fig 3: Light gathering capacity of an optical fibre.

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Optical fibre characterization:

Objective: To determine the numerical aperture of a given multimode optical fibre.

Apparatus to be used: Diode laser, Optical laser, Laser-fibre coupler, Optical rail, Pinhole photo detector, Power supply for laser and detector output measurement unit.

Basic Theory:

- Optical fibres are optical waveguides used for data transmissions. They are made of low loss materials like glass and plastic. They have a central core through which the light is guided, embedded in an outer cladding of a slightly lower refractive index than the core. (See fig 1.)

- The geometry of launching a light ray into an optical fibre is shown in Fig 2, which illustrates a meridional ray A at the critical angle ϕ_c within the fibre at the core-cladding interface. It is observed that this ray enters the fibre core at angle θ_a to the fibre axis and is refracted at the air core interface before transmission to the core-cladding interface at the critical angle. Hence, any rays incident into the fibre core at an angle greater than θ_a will be transmitted to the core-cladding interface at an angle less than ϕ_c and will not totally internally reflected. This situation is also illustrated in fig 2, where the

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Observation table : least count = 0.01 mm

TR = MSR + HSR

HSR = HSC × LC

Z (mm)	Micrometer Reading (mm)			Output current (μA)	R = D/2 (mm)
	MSR	HSR	TR		
1	16	0.06	16.06	0.0	0.8
1	16.5	0.20	16.70	20.3	
1	16.5	0.30	16.80	40.3	
1	16.5	0.38	16.88	60.2	
1	16.5	0.44	16.94	80.3	
1	17	0	17.00	100.6	
1	17	0.04	17.04	120.4	
1	17	0.07	17.07	140.5	
1	17	0.12	17.12	160.5	
1	17	0.21	17.21	180.3	
1	17	0.27	17.27	184.0	
1	17	0.37	17.37	160.3	
1	17	0.43	17.43	140.2	
1	17	0.46	17.46	120.5	
1	17.5	0.02	17.52	100.5	

From the graph, Beam radius corresponding to 5% of maximum attainable intensity = 0.8 mm

- Distance between detector and fibre optic end (z) = 1 mm.
- Acceptance angle = 38.65°
- Numerical angle = 0.624

1	17.5	0.06	17.56	80.5
1	17.5	0.13	17.62	60.2
1	17.5	0.22	17.72	40.1
1	17.5	0.32	17.82	20.2
1	18.5	0.13	18.63	0.0

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incident ray B at an angle greater than θ_a is refracted into the cladding and eventually lost by radiation. Thus for rays to be transmitted by total internal reflection within the fibre core, they must be incident on the fibre core within an acceptance cone defined by the conical half angle θ_a. Where θ_a, is the maximum angle to the axis at which light may enter the fibre to be propogated, and is referred to as the acceptance angle for the fibre.

- Numerical aperature (NA) is the light gathering capacity of an optical fibre. The NA is a measure of ~~how~~ how much light can be collected by an optical system. NA varies according to an angle known as the acceptance angle θ_a, which gives the size of a cone of light that can be accepted by the fibre.

- From fig 3 acceptance angle can be defined as:

$$\theta_a = \tan^{-1}\left(\frac{R}{Z}\right)$$

R is the radius corresponding to 5% of the maximum attainable intensity.
Z is the distance between fibre output end and the detector.

- The sine of the acceptance angle is defined as numerical aperture (NA). Therefore,

$$NA = \sin\theta_a$$

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From graph:

$D = 18 - 16.4$

$= 1.6$

$R = \frac{1.6}{2} = 0.8\ mm$

$\therefore \theta_a = \tan^{-1}(0.8)$

$= 38.65$

$NA = \sin\theta_a$

$= \sin(38.65)$

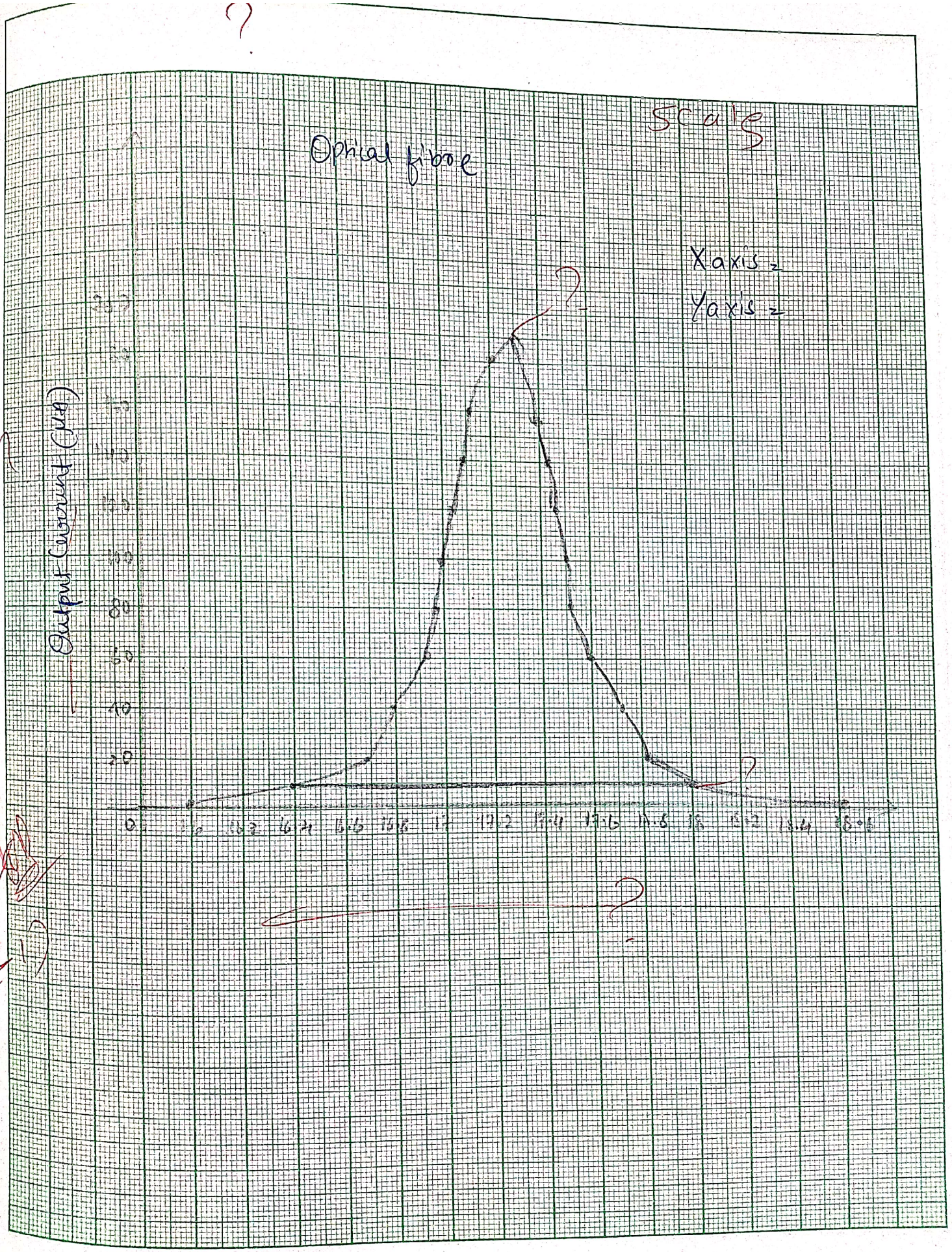
$= 0.624$

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Result: The Numerical aperture (NA) of given multimode fibre: 0.624 .


Optical fibre
Scale
X axis =
Y axis =
Output Current (μA)